Analytical Chemistry Cumulative Exam
Spectroscopy (Unannounced)
March 8, 2019

1. (40 points) Spectroscopy is an integral part to analytical chemistry and requires some understanding of physical chemical principals. For each of the types of spectroscopy listed below, draw and fully label a Jablonski (energy) diagram illustrating the chemical or physical process that is occurring. Please use a different diagram for each process (6 diagrams). Briefly comment about the similarities and differences between these different types of spectroscopy.

a) UV-Vis

b) Fluorescence

c) Phosphorescence

d) Atomic Absorption

e) Infrared

f) Raman

2. (60 points) For the types of spectroscopy listed below, use sketches and brief descriptions to compare and contrast the different instruments that are used to obtain these spectra. Highlight the similarities and differences among these instruments. Be sure to indicate the type of light sources used and how wavelength selection and noise reduction occurs within the instruments.

a) UV-Vis

b) Fluorescence

c) Atomic Absorption

d) Infrared

e) Raman